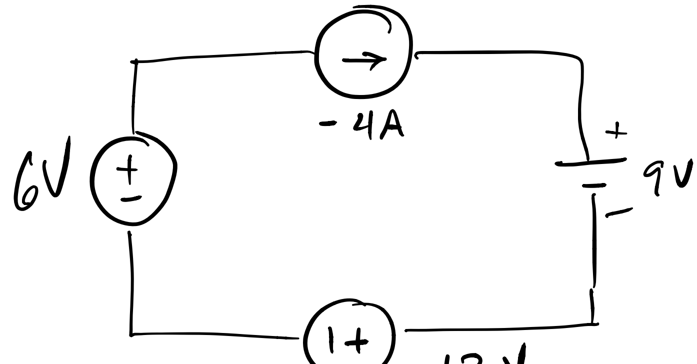
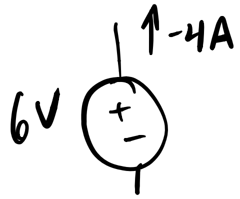
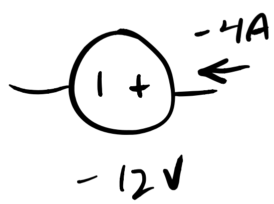


$$P_{6V} = \overset{\text{rise}}{-}\, V I = -6 \cdot -4 = 24\,W$$

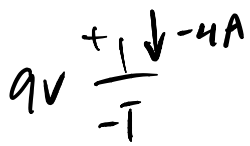
(dissipating)





$$P_{-12V} = +V I = -12 \cdot -4 = 48\,W$$

(diss.)



$$P_{9V} = +V I = +9 \cdot -4 = -36\,W$$

(or 36W gen.)

by Conservation of Power:

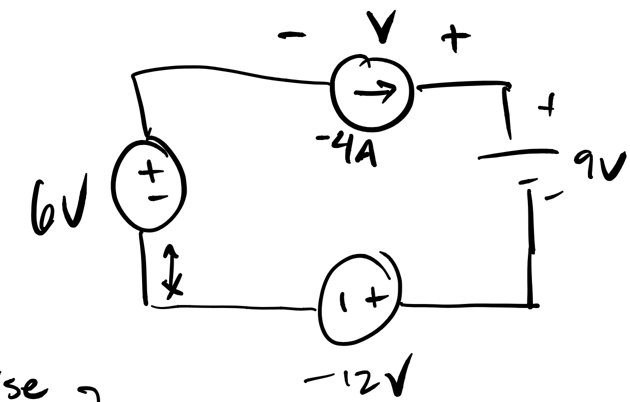
watch signs!!!

$$\sum P = P_{6V} + P_{-12V} + P_{9V} + P_{4A} = 0$$

$$P_{4A} = -P_{6V} - P_{-12V} - P_{9V} = -24 - 48 - -36$$

$P_{-4A} = -36\,W$ or 36W generating

KVL :   CW, LLH



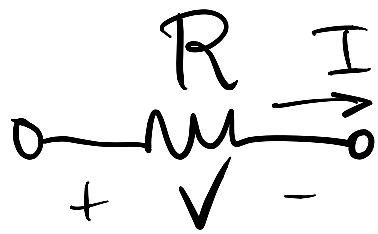
rise   drop

$$-6 - V + 9 + (-12) = 0$$

$$V = -6 + 9 - 12 = -9V$$

$$P_{-4A} = -V \cdot I = -(-9) \cdot -4 = -36W$$

Resistors



- two-terminal device that opposes the flow of current
- unit: ohm (Ω)

Ohm's Law:

$$R = \frac{V}{I} \text{ or } V = IR \text{ or } I = \frac{V}{R}$$

∴ resistors can <u>only</u> dissipate power!

<u>Passive</u> device

∴ voltage always <u>drops</u> in the direction of positive current!!!

power in a resistor:

$$P = VI$$

$$\therefore P = I \cdot R \cdot I = I^2R$$

$$\text{OR } P = V \cdot \frac{V}{R} = \frac{V^2}{R}$$

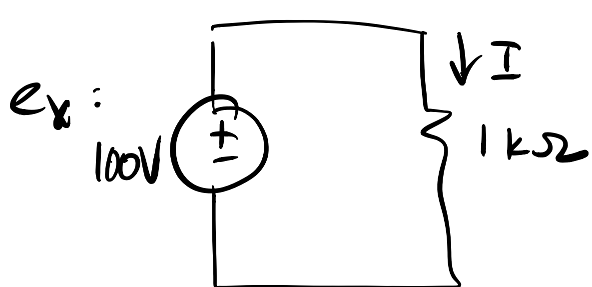
∴ physical resistors have a resistance and <u>power rating</u>.

∴ small resistors used in electronics:

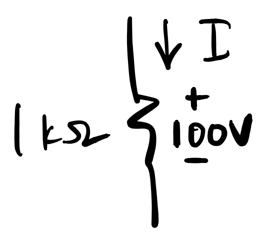
1/8 W, 1/4 W, 1/2 W, 1W, 2W, ↑

- small carbon film resistors in lab: 1/4 W, 5% tolerance
- significance of power rating is that resistors dissipate power by giving off <u>heat</u>.
  - power rating is just as important as value!
  - for long life and reliability, derate (use higher power rating than necessary)

factor of three → good idea!



- determine I and power dissipated in resistor



← we know this voltage because they're in parallel

$$I = \frac{V}{R} = \frac{100V}{1k} = 100\,mA$$

- if circuit is entirely in kΩ and V, currents will be in mA → save steps, but be careful!

G	giga	10^9
M	mega	10^6
k	kilo	10^3
m	milli	10^{-3}
μ	micro	10^{-6}
n	nano	10^{-9}
p	pico	10^{-12}
f	femto	10^{-15}

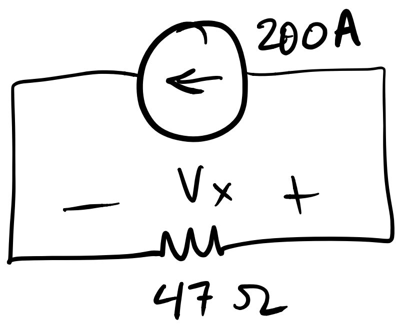
$$P = \frac{V^2}{R}$$

$$= \frac{100^2}{1k}$$

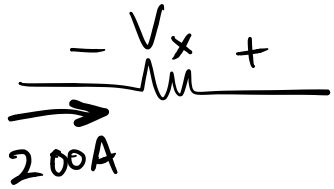
$$= 10{,}000 \text{ mW}$$

OR 10 W

ex:



.. determine V_x and the power in the resistor



rise!

$$-V_x = IR$$

$$V_x = -IR$$

$$= -200 \cdot 47$$

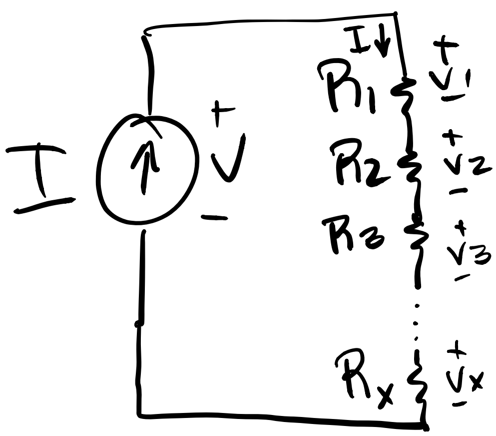
$$V_x = -9400 \text{ V}$$

or -9.4 kV

$$P = I^2 R = 200^2 \cdot 47 = 1.88 \text{ MW}$$

Resistors in Series

∴ consider a number of resistors in series, connected to a current source



∴ We know by Ohm's Law:

$$V_1 = I R_1$$
$$V_2 = I R_2$$
etc..

∴ we know by KVL:

$$-V + V_1 + V_2 + \ldots + V_x = 0$$
$$V = IR_1 + IR_2 + \ldots + I R_x$$

.. the equivalent resistance "seen" by the current source is

$$R = \frac{V}{I} = \frac{\cancel{I}R_1 + \cancel{I}R_2 + \ldots + \cancel{I}R_x}{\cancel{I}}$$

$$R_{series} = R_1 + R_2 + \ldots + R_x$$

.. resistors in series add

.. if all equal-value resistors:

$$R_{series} = nR$$

.. the reciprocal of resistance is conductance

Homework: read S&M 2.4-2.8

G

unit: siemens

or ohm^{-1}

or mho ($\mho$)

upside-down omega!!!

$$G = \frac{1}{R}$$

$$\therefore G_{series} = \frac{1}{\frac{1}{G_1} + \frac{1}{G_2} + \ldots + \frac{1}{G_x}}$$